

Natural Language Processing

Natural language processing (NLP) is the processing of natural language information by a computer. NLP is a subfield of computer science and is closely associated with artificial intelligence. NLP is also related to information retrieval, knowledge representation, computational linguistics, and linguistics more broadly.

Major processing tasks in an NLP system include: speech recognition, text classification, natural language understanding, and natural language generation.

== History ==

Natural language processing has its roots in the 1950s. Already in 1950, Alan Turing published an article titled "Computing Machinery and Intelligence," which proposed what is now called the Turing test as a criterion of intelligence, though at the time that was not articulated as a problem separate from artificial intelligence. The proposed test includes a task that involves the automated interpretation and generation of natural language.

=== Symbolic NLP (1950s ? early 1990s) ===

The premise of symbolic NLP is often illustrated using John Searle's Chinese room thought experiment: Given a collection of rules (e.g., a Chinese phrasebook, with questions and matching answers), the computer emulates natural language understanding (or other NLP tasks) by applying those rules to the data it confronts.

1950s: The Georgetown experiment in 1954 involved fully automatic translation of more than sixty Russian sentences into English. The authors claimed that within three or five years, machine translation would be a solved problem. However, real progress was much slower, and after the ALPAC report in 1966, which found that ten years of research had failed to fulfill the expectations, funding for machine translation was dramatically reduced. Little further research in machine translation was conducted in America (though some research continued elsewhere, such as Japan and Europe) until the late 1980s when the first statistical machine translation systems were developed.

1960s: Some notably successful natural language processing systems developed in the 1960s were SHRDLU, a natural language system working in restricted "blocks worlds" with restricted vocabularies, and ELIZA, a simulation of Rogerian psychotherapy, written by Joseph Weizenbaum between 1964 and 1966. Despite using minimal information about human thought or emotion, ELIZA was able to produce interactions that appeared human-like. When the "patient" exceeded the very small knowledge base, ELIZA might provide a generic response, for example, responding to "My head hurts" with "Why do you say your head hurts?". Ross Quillian's successful work on natural language was demonstrated with a vocabulary of only twenty words, because that was all that would fit in a computer memory at the time.

1970s: During the 1970s, many programmers began to write "conceptual ontologies", which structured real-world information into computer-understandable data. Examples are MARGIE (Schank, 1975), SAM (Cullingford, 1978), PAM (Wilensky, 1978), TaleSpin (Meehan, 1976), QUALM (Lehnert, 1977), Politics (Carbonell, 1979), and Plot Units (Lehnert 1981). During this time, the first chatterbots were written (e.g., PARRY).

1980s: The 1980s and early 1990s mark the heyday of symbolic methods in NLP. Focus areas of the time included research on rule-based parsing (e.g., the development of HPSG as a computational operationalization of generative grammar), morphology (e.g., two-level morphology), semantics (e.g., Lesk algorithm), reference (e.g., within Centering Theory) and other areas of natural

language understanding (e.g., in the Rhetorical Structure Theory). Other lines of research were continued, e.g., the development of chatterbots with Racter and Jabberwacky. An important development (that eventually led to the statistical turn in the 1990s) was the rising importance of quantitative evaluation in this period.

=== Statistical NLP (1990s?present) ===

Up until the 1980s, most natural language processing systems were based on complex sets of hand-written rules. Starting in the late 1980s, however, there was a revolution in natural language processing with the introduction of machine learning algorithms for language processing. This shift was influenced by increasing computational power (see Moore's law) and a decline in the dominance of Chomskyan linguistic theories (e.g. transformational grammar), whose theoretical underpinnings discouraged the sort of corpus linguistics that underlies the machine-learning approach to language processing.

1990s: Many of the notable early successes in statistical methods in NLP occurred in the field of machine translation, due especially to work at IBM Research, such as IBM alignment models. These systems were able to take advantage of existing multilingual textual corpora that had been produced by the Parliament of Canada and the European Union as a result of laws calling for the translation of all governmental proceedings into all official languages of the co